MARSEL BAKASHOV

AI & Software Engineering Student
Data Science / ML / RL / AI

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ABOUT ME

Software Engineering student specializing in Artificial Intelligence, with expertise in Machine Learning, Reinforcement Learning, and Agentic AI Systems. Experienced in applying AI to real-world industrial challenges, including dynamic pricing, predictive maintenance, semantic search, and decision-support workflow automation.

SKILLS

Programming: Python, C++, Java, SQL, Bash, R
ML / Data: PyTorch, Scikit-learn, TensorFlow, HuggingFace, MLflow
Tools: Docker, Git, Airflow, Azure, Power BI, Jenkins, Databricks, Snowflake
Languages: English, French, Russian, Spanish, Korean

EDUCATION

1/2026 – 2/2027 **Professional Master's in Machine Learning** — ***University of Montreal / McGill - MILA***
GPA: 3.4 / 4.0 — Montreal, Canada

9/2021 – 12/2025 **Bachelor of Software Engineering** — ***Polytechnique Montreal***
GPA: 3.4 / 4.0 — Montreal, Canada

2/2025 – 6/2025 **Bachelor of Computer Engineering** — ***Korea University***
Academic Exchange — Seoul, South Korea

EXPERIENCE

9/2023 – 12/2023 **Data Science & ML Intern** — ***McKesson***
• Developed AI agents and automated workflows to analyze business data, generate executive summaries, and support decision-making.
`Power BI / Power Automate / Python / Azure / Claude Code / Snowflake`

1/2026 – 5/2026 **Data Scientist & RL Intern** — ***Intact Insurance***
• Developed constrained RL pricing policies using adaptive primal–dual methods, validated through simulation and risk metrics, increasing long-term profitability while lowering volatility.
`Python / PyTorch / RL / CVar / MLflow / Databricks / Snowflake / Risk Metrics`

6/2025 – 9/2025 **Data Scientist & ML Intern** — ***Pratt & Whitney***
• Developed predictive models to estimate engine remaining useful life, enabling better maintenance planning and reducing unplanned downtime by 15%.
`Python / Scikit-learn / PyTorch / Pandas / NumPy / MLflow / Jupyter`

5/2024 – 9/2024 **Data Scientist Intern** — ***Radio-Canada / CBC***
• Designed an AI-powered indexing system for semantic and keyword search (100+ employees).
`ELK / ETL / Airflow / Python / Azure AI Search / Hugging Face / Embeddings`

PROJECTS

Personal Project **GravityGuyML — Reinforcement Learning Side-Scroller** — ***GitHub***
• Built a Gymnasium-compatible Pygame environment with procedural generation and deep RL agents (`DQN`, `PPO`, `SAC`, `A2C`).
`Python / Machine Learning / Deep Learning / Reinforcement Learning / TensorFlow`

Personal Project **Quantitative Finance Notes — Monte Carlo & GBM Modeling** — ***GitHub***
• Built reproducible notebooks on stochastic processes, Monte Carlo simulations, and Geometric Brownian Motion for market modeling and data calibration.
`Python / NumPy / Pandas / Matplotlib / Monte Carlo / Quantitative Finance / GBM`

Academic Project **Weather-Based Disease Outbreak Prediction** — ***GitHub***
• Predicted outbreaks from climate data using classification models (Logistic Regression, SVM, Decision Trees) with F1-score and AUC-ROC evaluation.
`Python / Jupyter / pandas / scikit-learn / Matplotlib / FP-Growth`

EXTRACURRICULAR ACTIVITIES

1/2024 – 8/2024 **Teaching Assistant – Discrete Mathematics** — ***Polytechnique Montreal***
• Led review sessions, graded assignments, and supported 80+ students.

1/2024 – 5/2024 **Sponsorship Director** — ***PolyAI Hackathon***
• Raised over $22,000 in funding and led a sponsorship team with weekly meetings.